

Impulse and Impulsive Force

§ Impulse of a force:-

The impulse of a force is defined to be product of the force and time during which the force acts on a mass, when the force is constant in magnitude and direction.

II 1. $\boxed{\text{Impulse (I)} = P \times t}$, $P = \text{force}$ and $t = \text{time duration}$

2. $\text{Impulse (I)} = m(v - u)$

$\boxed{\text{Impulse} = \text{change in momentum}}$

3. $\boxed{\text{Work done} = I \cdot \frac{1}{2}(u+v)}$

§ Impulsive force:- An impulsive force is a very large force of an extremely short duration, such that impulse of the force i.e. the change in momentum produced by it in the body, is finite, but the displacement of the body during the short interval is negligible.

§ Principle of conservation of linear momentum:-

If the sum of the external forces acting on a system of bodies be zero in any direction, then the total momentum of the system in that direction remains constant during the motion.

& Problem 1:- A shot of mass 40 kg leaves a gun with a velocity of 1000 cm/sec. If the velocity of recoil is 10 cm/sec. Find the mass of the gun.

Ans. Let the mass of the gun be M kg.

By the principle of linear momentum

$$M \times 10 \times 1000 = 40 \times 1000 \times 1000$$

$$\text{or } M = 4000 \text{ kg.}$$

& Problem 2:- A force acting on a body of mass 5 lb changes its velocity from 30 m.p.h. to 45 m.p.h. Find the impulse of the force.

Ans. Impulse of the force (P) = $m(v - u)$

$$= 5 \times \frac{(45 - 30)}{60 \times 60} \times 1760 \times 3 \text{ poundal sec.}$$

$$= 110 \text{ poundal sec.}$$

& Problem 3:- A mass of 5 lbs attached to a string is at rest. If a sudden jerk of the string produces in the mass a velocity of 12 ft/sec. Find the impulse of the jerk.

Ans. Impulse of the jerk

= Change in momentum

$$= m(v - u)$$

$$= 5(12 - 0) \text{ Poundal sec.}$$

$$= 60 \text{ Poundal sec.}$$

& Problem 4:- When the gun powder bursts inside a gun of mass 10 lbs, the bullet of mass 1 ounce is projected with a velocity of 1000 ft/sec. What is the kinetic energy generated by the explosion.

Ans. Here, mass of the bullet = 1 ~~ounce~~ ounce = $\frac{1}{16}$ lbs.

If v is the velocity of recoil of the gun.

Then by the principle of conservation

$$10v = \frac{1}{16} \times 1000$$

$$\therefore v = \frac{25}{4} \text{ ft/sec.}$$

If the total kinetic energy generated.

$$E = \text{K.E. of the gun} + \text{K.E. of the bullet,}$$

$$= \frac{1}{2} \times 10 \times \left(\frac{25}{4}\right)^2 + \frac{1}{2} \times \frac{1}{16} \times (1000)^2 \text{ ft Poundal}$$

$$= 31445.31 \text{ ft Poundal}$$

3) Problem 5: A cricket ball moving with a velocity of 25 m/s is struck by a bat which causes it to move in the same straight line but in opposite direction with a velocity of 15 m/s . If the impulse generated by the bat is 600 N-s . Find the mass of the ball.

Ans. Here $u = 25 \text{ m/s} = 2500 \text{ cm/s}$.

and $v = 15 \text{ m/s} = 1500 \text{ cm/s}$ but in the direction opposite to that of u .

Let m be the mass of the ball.

Here $I = 600 \text{ N-s}$.

$$\text{Now } I = m(v - (-u))$$

$$600 = m(u + v)$$

$$= m(1500 + 2500)$$

$$\Rightarrow m = \frac{600}{4000} = \frac{3}{20} \text{ gm.}$$

3) Problem 6: A shell of mass m is projected from a gun of mass M by an explosion which generates kinetic energy E . Prove that the initial velocity of the shell is $\sqrt{\frac{2ME}{(M+m)m}}$.

Ans. Let v_1 be the initial velocity of the shell and v_2 be the velocity of mass M .

By the principle of conservation of linear momentum

$$mv_1 = Mv_2 \quad \text{--- (1)}$$

Now the K.E. generated in the system for gun as well as the shell together is

$$E = \frac{1}{2}mv_1^2 + \frac{1}{2}Mv_2^2$$

$$E = \frac{1}{2}mv_1^2 + \frac{1}{2}Mv_2^2$$

$$= \frac{1}{2}m \left[v_1 + \left(\frac{mv_1}{M} \right) \right]^2$$

$$= \frac{1}{2} \left(mv_1^2 + M \left(\frac{mv_1}{M} \right)^2 \right)$$

$$2E = m \left[Mv_1^2 + m^2v_1^2 \right]$$

$$= \frac{1}{2} \frac{m}{M} (Mv_1^2 + mv_1^2)$$

$$\Rightarrow v_1^2 = \frac{2EM}{m(M+m)}$$

$$\Rightarrow 2EM = m v_1^2 (M+m)$$

$$\Rightarrow v_1^2 = \frac{2EM}{m(M+m)}$$

$$\Rightarrow v_1 = \sqrt{\frac{2EM}{m(M+m)}}$$

* ~~Ans~~

8 Problem 7: A body of mass $(m_1 + m_2)$ split into two parts of masses m_1 and m_2 by an internal explosion which generates kinetic energy E . Show that if after explosion the parts move in the same line as before, their relative speed is $\sqrt{\frac{2E(m_1 + m_2)}{m_1 m_2}}$.

Ans. Let v be the Velocity of the body before explosion and v_1 & v_2 be the Velocities of two parts of the body m_1 and m_2 after explosion respectively.

Now by the principle of conservation of linear momentum we get,

$$(m_1 + m_2)v = m_1 v_1 + m_2 v_2 \quad \text{--- (1)}$$

After explosion kinetic energy generates,

$$\therefore E = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 - \frac{1}{2} (m_1 + m_2) v^2$$

$$2E = m_1 v_1^2 + m_2 v_2^2 - (m_1 + m_2) \left\{ \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} \right\}^2$$

$$\therefore 2E(m_1 + m_2) = (m_1 + m_2) (m_1 v_1^2 + m_2 v_2^2) - (m_1 v_1 + m_2 v_2)^2$$

$$= m_1 m_2 (v_1^2 + v_2^2 - 2v_1 v_2)$$

$$\therefore (v_1 - v_2)^2 = \frac{2E(m_1 + m_2)}{m_1 m_2}$$

$$\therefore v_1 - v_2 = \sqrt{\frac{2E(m_1 + m_2)}{m_1 m_2}}$$

9 Problem 8: A gun of mass M fires a shell of mass m horizontally and the energy of explosion is such as would be sufficient to project the shell vertically to a height h . Show that the velocity of recoil of the gun is $\left\{ \frac{2mgh}{M(m+M)} \right\}^{1/2}$.

Ans. Let E be the energy generated by the explosion.

Since the energy is just sufficient to raise a mass m to a height h . $E = mgh$ --- (1)

If u be the velocity of the shell and v that of recoil of the gun.

$$mu = Mv$$

$$\therefore u = \frac{Mv}{m} \quad \text{--- (2)}$$

$$\text{Energy of explosion} = E = \frac{1}{2} mu^2 + \frac{1}{2} Mv^2$$

$$\therefore mgh = \frac{1}{2} m \frac{M^2 v^2}{m^2} + \frac{1}{2} Mv^2$$

$$\alpha, 2m\sqrt{gh} = Mv\sqrt{(M+m)}$$

$$\alpha, v = \frac{2m\sqrt{gh}}{M(M+m)}$$

$$\alpha v = \left\{ \frac{2m\sqrt{gh}}{M(M+m)} \right\}^{\frac{1}{2}}$$

3 Problem 9: A bullet of mass m moving with a velocity u strikes a block of mass M which is free to move in the direction of motion of the bullet and is embedded in it. Show that the loss of K.E is

$$\frac{1}{2} \cdot \frac{mM}{m+M} \cdot u^2$$

Ans. If v be the velocity of the block of mass M after the bullet is embedded in it, then by the principle of conservation of linear momentum

$$(m+M)v = mu$$

$$\alpha, v = \frac{mu}{m+M} \quad \text{--- (1)}$$

Then loss of K.E = Initial K.E - K.E after the bullet is embedded

$$= \frac{1}{2} mu^2 - \frac{1}{2} (m+M)v^2$$

$$= \frac{1}{2} mu^2 - \frac{1}{2} (m+M) \cdot \frac{m^2 u^2}{(m+M)^2} \quad \text{--- (by 1)}$$

$$= \frac{1}{2} mu^2 \left\{ 1 - \frac{m}{m+M} \right\}$$

$$= \frac{1}{2} \cdot \frac{mMu^2}{m+M}$$

3 Problem 10: - A shell of mass M is moving with velocity v . An internal explosion generates an amount of energy E and breaks the shell into two fragments whose masses are in the ratio of $m_1 : m_2$. If the fragments continue to move in the original direction of motion, show that their velocities are

$$v + \sqrt{\frac{2m_2 E}{m_1 M}} \quad , \quad v - \sqrt{\frac{2m_1 E}{m_2 M}}$$

Ans. Let after explosion, the mass M breaks into two fragments, whose masses are M_1 and M_2 and v_1, v_2 be their respective velocities ($v_1 > v_2$).

From the given condition, derive

$$\frac{M_1}{M_2} = \frac{m_1}{m_2}$$

$$\text{or, } \frac{M_1}{m_1} = \frac{M_2}{m_2} = \frac{M_1 + M_2}{m_1 + m_2} = \frac{M}{m_1 + m_2}$$

$$\therefore M_1 = \frac{m_1 M}{m_1 + m_2} \quad \text{and} \quad M_2 = \frac{m_2 M}{m_1 + m_2} \quad \text{--- (1)}$$

Now K.E. before explosion + E = K.E. after explosion

$$\text{or, } \frac{1}{2} M V^2 + E = \frac{1}{2} M_1 v_1^2 + \frac{1}{2} M_2 v_2^2 \quad \text{--- (2)}$$

Now the principle of linear momentum.

$$M V = M_1 v_1 + M_2 v_2$$

$$\text{or } v_2 = \frac{M V - M_1 v_1}{M_2}$$

Putting the value of v_2 in (2), we get

$$\frac{1}{2} M V^2 + E = \frac{1}{2} M_1 v_1^2 + \frac{1}{2} M \left(\frac{M V - M_1 v_1}{M_2} \right)^2$$

$$\text{or, } M M_2 v^2 + 2 E M_2 = M_1 M_2 v_1^2 + M^2 v^2 + M_1^2 v_1^2 - 2 M M_1 v v_1$$

$$\text{or, } M (M - M_1) v^2 + 2 M_2 E = M_1 (M - M_1) v_1^2 + M^2 v^2 + M_1^2 v_1^2 - 2 M M_1 v v_1$$

$$[\text{as } M = M_1 + M_2]$$

$$\text{or, } M^2 v^2 - M M_1 v^2 + 2 M_2 E = M_1 M v_1^2 - M_1^2 v_1^2 + M^2 v^2 + M_1^2 v_1^2 - 2 M M_1 v v_1$$

$$\text{or, } M M_1 (v_1^2 - 2 v v_1 + v^2) = 2 M_2 E$$

$$\text{or, } M M_1 (v_1 - v)^2 = 2 M_2 E$$

$$\text{or, } (v_1 - v)^2 = \frac{2 M_2 E}{M M_1} = \frac{2 E}{M} \left(\frac{m_2}{m_1} \right) \quad \left[\text{Since } \frac{M_2}{M_1} = \frac{m_2}{m_1} \right]$$

$$\text{or, } v_1 - v = \pm \sqrt{\frac{2 m_2 E}{m_1 M}}$$

$$\text{or, } v_1 = v \pm \sqrt{\frac{2 m_2 E}{m_1 M}} = v + \sqrt{\frac{2 m_2 E}{m_1 M}} \quad (\text{Since } v_1 > v_2)$$

Similarly, eliminating v_2 from (1) and (2), we get,

$$(v_2 - v)^2 = \frac{2 m_1 E}{m_2 M}$$

$$\text{or } v_2 = v \pm \sqrt{\frac{2 m_1 E}{m_2 M}} = v - \sqrt{\frac{2 m_1 E}{m_2 M}} \quad (\text{Since } v_1 > v_2)$$

§ Problem 11:- A shell, lying in a straight smooth horizontal tube, suddenly explodes and breaks into portions of masses m_1 and m_2 . If D is the distance apart of the masses after a time t . Show that the work done by the explosion is

$$\frac{1}{2} \cdot \frac{m_1 \cdot m_2}{m_1 + m_2} \cdot \frac{D^2}{t^2}$$

Ans. Let v_1 and v_2 be the velocities after explosion of the portion m_1 and m_2 respectively in opposite directions along the tube. Now by the principle of conservation of momentum, we get

$$m_1 v_1 - m_2 v_2 = 0$$

$$\text{or } m_1 v_1 = m_2 v_2 \quad \text{--- (1)}$$

The distance apart between the portions after time t is

$$(v_1 + v_2)t = D \quad \text{--- (2)}$$

From (1), we get, $\frac{v_1}{m_2} = \frac{v_2}{m_1} = \frac{v_1 + v_2}{m_1 + m_2} = \frac{D}{t(m_1 + m_2)} \quad [By (2)]$

The work done by the explosion is the K.E. generated by it

$$= \frac{1}{2} (m_1 v_1^2 + m_2 v_2^2)$$

$$= \frac{1}{2} [m_1 m_2^2 + m_2 m_1^2] \left\{ \frac{D}{t(m_1 + m_2)} \right\}^2 \quad [By (3)]$$

$$= \frac{1}{2} m_1 m_2 [m_1 + m_2] \cdot \frac{D^2}{t^2 (m_1 + m_2)^2}$$

$$= \frac{1}{2} \cdot \frac{m_1 m_2}{m_1 + m_2} \cdot \frac{D^2}{t^2}$$

§ Problem 12:- For a rectilinear motion of a particle, if an impulse P changes its velocity from u to v and if E be the change of K.E., then show that

$$E = \frac{1}{2} P(u + v)$$

Ans. $E = \text{Change in K.E.}$

$= \text{Work done by the force}$

$$= P \Delta x$$

$$= m f \Delta x \quad [P = m f]$$

$$= m f (v^2 - u^2)$$

$$= m (v - u) \cdot f (u + v)$$

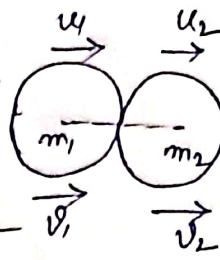
$$= P \cdot f (u + v)$$

$$\boxed{E = f \cdot P \cdot (u + v)}$$

§ Collision of Elastic Bodies §

§ Direct impact of two smooth spheres.

Ans. Let the smooth spheres of masses m_1 and m_2 impinge directly with velocity u_1 and on another smooth sphere of mass m_2 moving in the same direction with velocity u_2 .



Let v_1 and v_2 be the velocities of the two spheres after impact.

The velocity of approach is $(u_1 - u_2)$ and the velocity of separation is $(v_2 - v_1)$. Then from Newton's experimental law of impact, we get

$$v_2 - v_1 = e(u_1 - u_2) \quad \text{--- (1)}$$

where e is the co-efficient of restitution.

Again, since the impulsive action and reaction is acting along the line of centres, by the principle of conservation of momentum, we have

$$m_1 v_1 + m_2 v_2 = m_1 u_1 + m_2 u_2 \quad \text{--- (2)}$$

[$\therefore$ The total momentum along the line of impact is unaltered]

Multiplying (1) by m_2 and then subtracting from (2), we get

$$m_1 v_1 + m_2 v_2 = m_1 u_1 + m_2 u_2 - e m_2 u_1 + e m_2 u_2$$

$$\text{or, } (m_1 + m_2) v_1 = u_1 (m_1 - e m_2) + m_2 u_2 (1 + e) \quad \text{--- (3)}$$

Again multiplying (1) by m_1 and then adding to (2), we have

$$(m_1 + m_2) v_2 = m_1 u_1 (1 + e) + u_2 (m_2 - e m_1) \quad \text{--- (4)}$$

Equation (3) $\times$ (4) gives the velocities v_1 and v_2 of the two spheres just after impact.

Loss of kinetic energy due to impact

= K.E before impact - K.E after impact

$$= \frac{1}{2} (m_1 u_1^v + m_2 u_2^v) - \frac{1}{2} (m_1 v_1^v + m_2 v_2^v)$$

$$= \frac{1}{2} \cdot \frac{1}{m_1 + m_2} \left\{ (m_1 + m_2) (m_1 u_1^v + m_2 u_2^v) - (m_1 + m_2) (m_1 v_1^v + m_2 v_2^v) \right\}$$

$$= \frac{1}{2} \cdot \frac{1}{m_1 + m_2} \left[m_1 u_1^v + m_2 u_2^v + m_1 \cdot m_2 (u_1^v + u_2^v) - \{ m_1 v_1^v + m_2 v_2^v + m_1 \cdot m_2 (v_1^v + v_2^v) \} \right]$$

$$= \frac{1}{2} \cdot \frac{1}{m_1 + m_2} \left[(m_1 u_1 + m_2 u_2)^2 + m_1 m_2 (u_1 - u_2)^2 - \{ (m_1 v_1 + m_2 v_2)^2 + m_1 m_2 (v_1 - v_2)^2 \} \right]$$

$$= \frac{1}{2} \cdot \frac{1}{m_1 + m_2} \left[m_1 m_2 (u_1 - u_2)^2 - m_1 m_2 e^2 (u_1 - u_2)^2 \right] \quad [\text{as } m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \text{ by Q}]$$

$$= \frac{1}{2} \cdot \frac{1}{m_1 + m_2} m_1 m_2 (u_1 - u_2)^2 (1 - e^2)$$

$$= \frac{m_1 m_2}{2(m_1 + m_2)} (1 - e^2) (u_1 - u_2)^2$$

If $0 \leq e < 1$, there is always a loss of K.E. due to impact. If $e = 0$, the bodies are perfectly inelastic and the loss of K.E. = $\frac{m_1 m_2}{2(m_1 + m_2)} (u_1 - u_2)^2$.

If $e = 1$, the two bodies are perfectly elastic and the loss of K.E. = 0 i.e. K.E. remains unchanged after impact.

Impulsive Action:-

Impulsive action on m_2 due to impact

= the change of momentum of m_2 due to impact.

$$= m_2 v_2 - m_2 u_2$$

$$= m_2 (v_2 - u_2)$$

$$= \frac{m_2}{m_1 + m_2} \left[(m_1 + m_2) v_2 - (m_1 + m_2) u_2 \right]$$

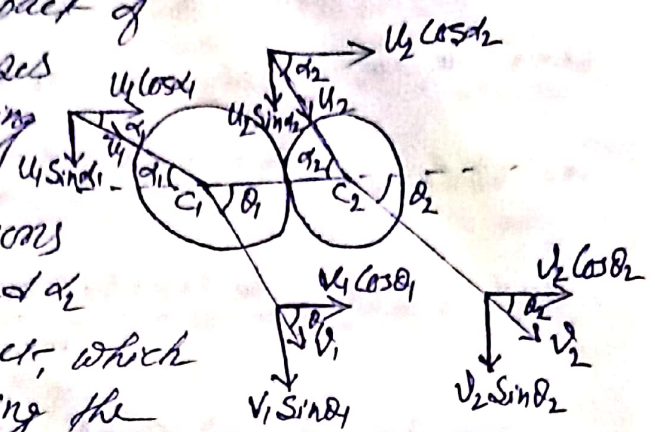
$$= \frac{m_2}{m_1 + m_2} \left[m_1 u_1 (1 + e) + u_2 (m_2 - e m_1) - (m_1 + m_2) u_2 \right] \quad \text{by Q}$$

$$= \frac{m_2}{m_1 + m_2} \left[m_1 u_1 (1 + e) - m_1 u_2 (1 + e) \right]$$

$$= \frac{m_1 m_2}{m_1 + m_2} (1 + e) (u_1 - u_2)$$

§ Oblique Impact of two smooth spheres:-

Ans. Let us consider the impact of two smooth spheres of masses m_1 and m_2 which are moving with velocities u_1 and u_2 respectively along directions which make angles α_1 and α_2 resp. to the line of impact, which is the straight line joining the centres C_1 and C_2 of the spherical masses.



Let the velocities of the spheres after impact be v_1 and v_2 along directions which make angles β_1 and β_2 resp. with the line of impact.

Since the spheres are smooth, the impulsive action and reaction between them will act along the line C_1C_2 , the common normal at the point of contact of the spheres. So, the components of velocities of the spheres along a line perpendicular to C_1C_2 will remain unaltered.

$$\text{So, } v_1 \sin \beta_1 = u_1 \sin \alpha_1 \quad \text{--- (1)}$$

$$\text{and } v_2 \sin \beta_2 = u_2 \sin \alpha_2 \quad \text{--- (2)}$$

By Newton's experimental law for relative velocities resolved along common normal at the point of contact.

We've

$$v_2 \cos \beta_2 - v_1 \cos \beta_1 = e (u_1 \cos \alpha_1 - u_2 \cos \alpha_2) \quad \text{--- (3)}$$

The impulsive forces which act during the collision of the spheres along their line of contact centres are equal and opposite. Thus total momentum remains unchanged.

$$\text{So, } m_1 v_1 \cos \beta_1 + m_2 v_2 \cos \beta_2 = m_1 u_1 \cos \alpha_1 + m_2 u_2 \cos \alpha_2 \quad \text{--- (4)}$$

from ③ and ④

$$(m_1 + m_2) v_1 \cos \theta_1 = (m_1 - e m_2) u_1 \cos \alpha_1 + m_2 (1+e) u_2 \cos \alpha_2$$

$$\text{and } (m_1 + m_2) v_2 \cos \theta_2 = m_1 (1+e) u_1 \cos \alpha_1 + (m_2 - e m_1) u_2 \cos \alpha_2$$

$$\text{or, } v_1 \cos \theta_1 = \frac{(m_1 - e m_2) u_1 \cos \alpha_1 + m_2 (1+e) u_2 \cos \alpha_2}{m_1 + m_2} \quad \text{--- ⑤}$$

$$\text{and, } v_2 \cos \theta_2 = \frac{m_1 (1+e) u_1 \cos \alpha_1 + (m_2 - e m_1) u_2 \cos \alpha_2}{m_1 + m_2} \quad \text{--- ⑥}$$

Squaring ⑤ and ⑥ and adding we get v_1^2 and v_2^2 .
Similarly from ② and ③, we get $\tan \theta_1$ and $\tan \theta_2$.

Similarly from ② and ③, we get v_2^2 and $\tan \theta_2$.

Impulsive Action and Reaction:-

The impulse of the blow being measured by the change of momentum produced along the line of impact, we are to take into consideration only the components of initial and final velocities of the spheres along this line and the required impulsive blow is given by

$$P = m_2 (v_2 \cos \theta_2 - u_2 \cos \alpha_2)$$

$$= \frac{m_1 m_2}{m_1 + m_2} (1+e) (u_1 \cos \alpha_1 - u_2 \cos \alpha_2)$$

§ Loss of K.E:-

The loss of K.E. of the spheres by the impact

$$= \frac{1}{2} (m_1 u_1^2 + m_2 u_2^2) - \frac{1}{2} (m_1 v_1^2 + m_2 v_2^2)$$

$$= \frac{1}{2} \{ m_1 u_1^2 (\cos^2 \alpha_1 + \sin^2 \alpha_1) + m_2 u_2^2 (\cos^2 \alpha_2 + \sin^2 \alpha_2) \}$$

$$- \frac{1}{2} \{ m_1 v_1^2 (\cos^2 \theta_1 + \sin^2 \theta_1) + m_2 v_2^2 (\cos^2 \theta_2 + \sin^2 \theta_2) \}$$

$$= \frac{1}{2} \{ m_1 u_1^2 \cos^2 \alpha_1 + m_2 u_2^2 \cos^2 \alpha_2 \}$$

$$- \frac{1}{2} \{ m_1 v_1^2 \cos^2 \theta_1 + m_2 v_2^2 \cos^2 \theta_2 \} \quad [\text{by ⑤ \& ⑥}]$$

$$= \frac{1}{2} \cdot \frac{m_1 m_2}{m_1 + m_2} (1-e^2) (u_1 \cos \alpha_1 - u_2 \cos \alpha_2)^2$$

which is essentially positive since $e < 1$

Now with the help of ③ and ④ and proceeding exactly as in the case of direct impact

§ From what height must a heavy elastic ball be dropped on the floor so that rebounding on it it will reach a height of 8 m. ? $e = \frac{1}{2}$

Ans. Let the ball be dropped from a height of h , hit the floor with a velocity of u .

$$\text{Then } u^2 = 2gh$$

If v be the velocity of the ball just after impact with the floor.

$$v = eu$$

So, the height attained by the ball after first rebound

$$= \frac{v^2}{2g} = \frac{e^2 u^2}{2g} = \frac{e^2 2gh}{2g} = e^2 h = \left(\frac{1}{2}\right)^2 \cdot h = \frac{h}{4}$$

By the problem

$$\frac{h}{4} = 8$$

$$\Rightarrow h = 32$$

$\therefore$ It will reach a height 8 if it falls from 32 m.

3. A smooth sphere falling vertically from a height h impinges on a fixed smooth horizontal plane and rebounds to a height h_1 . Show that $h_1 = e^2 h$, where e is the co-efficient of restitution between the sphere and the plane.

Ans. Let u be the velocity of the sphere just before impact and v its velocity just after impact.

$$\text{Then } u^2 = 2gh \text{ and } v = eu, \quad v^2 = 2gh_1$$

$$\text{Now } v^2 = e^2 u^2$$

$$\Rightarrow e^2 u^2 = 2gh_1$$

$$\Rightarrow e^2 \times 2gh = 2gh_1$$

$$\Rightarrow \boxed{e^2 h = h_1}$$



§9 Prob: 1. Two equal elastic spheres each of mass m move in the same direction with equal velocities u and collide directly. Show that they interchange their velocities.

Ans. Let v_1 and v_2 be the velocities of the spheres after impact. By Newton's law on impact,

$$v_2 - v_1 = e(u - u) = 0$$

$$\Rightarrow v_2 = v_1$$

Also, by the principle of conservation of linear momentum,

$$m v_1 + m v_2 = m u + m u$$

$$\Rightarrow v_1 = v_2 = u$$

$\therefore$ So, the velocities are interchanged.

Prob: 4:- Two elastic spheres impinge directly with equal and opposite velocities. Find the ratio of their masses so that one of these may come to rest after impact, the ~~co~~ coefficient of restitution being e .